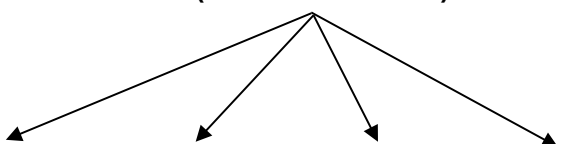


Structure of relational databases

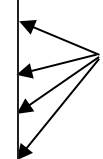
- Relational database is a collection of **tables**, each with a **unique name**
 - Example of table that stores information about instructors, identified by their *ID*
- Row in the table is a **relationship** between a set of values
 - In the example: relationship between an *ID* and the corresponding values for *name*, *dept_name*, and *salary* values
- Table is therefore collection of relationships
 - Connection between concepts of table and mathematical concept of relation (hence the name)
- In math terminology, a **tuple** is sequence (or list) of values
 - A relationship between n values is represented by an *n-tuple* of values (a tuple with n values)
 - This corresponds to a row in a table

attributes
(or columns)



<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

tuples
(or rows)



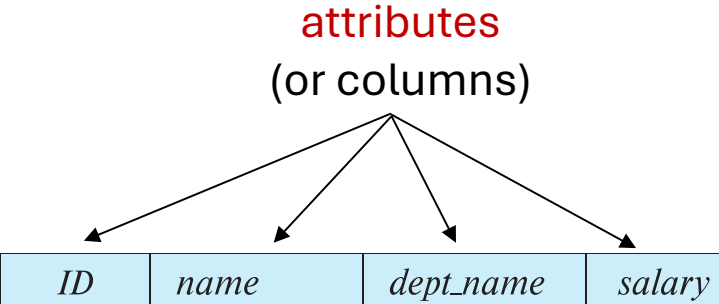
Relation Schema and Instance

- $A_1, A_2, \dots, A_n$ are **attributes**
- $R(A_1, A_2, \dots, A_n)$ is a **relation schema**

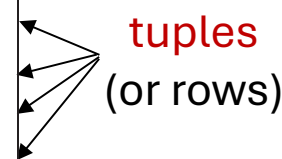
Example:

instructor (*ID*, *name*, *dept_name*, *salary*)

- A relation instance r defined over schema R is denoted by $r(R)$
 - The current values of a relation are specified by a table
- An element t of relation r is called a **tuple** and is represented by a row in a table



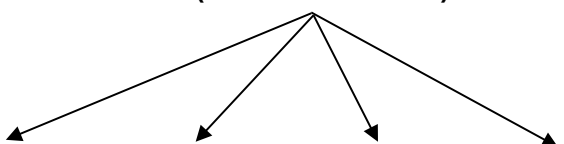
<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000



Attributes

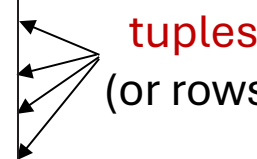
- The set of allowed values for each attribute is called the **domain** of the attribute
 - e.g., grades can be positive integers
- Attribute values are (normally) required to be **atomic**, that is, indivisible
 - e.g., consider a list of phone numbers
 - This depends on how we use domain elements, e.g., using the country code and area code and number separately
- The special value **null** is a member of every domain. Indicated that the value is “unknown”
 - *null* might value causes complications in the definition of many operations

attributes
(or columns)



<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

tuples
(or rows)



Relations are unordered

- Order of tuples is **irrelevant** (tuples may be stored in an arbitrary order)
- Example: instructor relations with ordered and unordered tuples

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

Database Schema

- **Database schema**: logical structure of the database
- **Database instance**: snapshot of the data in the database at a given instant in time
- Like variables in programming languages
 - Value of variable can change over time (contents of an instance can be updated)
 - Schema usually does not change

- Example:

- schema: *instructor (ID, name, dept_name, salary)*
- Instance:

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

Relating tuples of distinct relations

- Let's add the schema
department (*dept_name*, *building*, *budget*)
instructor (*ID*, *name*, *dept_name*, *salary*)
- Attribute *dept_name* appears in both schemas
 - Not a coincidence: common attributes in schemas is one way of relating tuples of distinct relations
- Example: find the information about all instructors that work in the Watson building
 - Start from *department* relation to find *dept_names* housed in Watson
 - For each such *dept_name*, we check *instructor* relation to find instructors affiliated with that *dept_name*
- Could we use pointers?
 - Relational model is based on values of domains
 - Advantages: independence from physical structure, records only relevant info, ease of portability, bi-directionality

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

<i>dept_name</i>	<i>building</i>	<i>budget</i>
Biology	Watson	90000
Comp. Sci.	Taylor	100000
Elec. Eng.	Taylor	85000
Finance	Painter	120000
History	Painter	50000
Music	Packard	80000
Physics	Watson	70000